

Bingzhuo TIAN

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B.S. candidate in Applied Physics with hands-on experience in embedded robotics, automated experimental systems, smart-car hardware, and open-source systems practice. Current interests: operating systems, embedded systems, robotics system software, ROS 2/DDS, real-time control-chain latency, device communication, and system observability.

EDUCATION

Hebei University of Technology | B.S. Candidate, Applied Physics 2023.08 – 2027.07

- GPA 3.18/4.00; 128.5 credits earned. Training in physics experiments, computational methods, fundamentals of electronic circuits, computational physics, and academic writing.

Hebei University of Technology | Micro-major, Complex Equipment Digital Twin 2024.11 – 2026.05

- GPA 3.64/4.00; key coursework: PLC Programming Practice (99/100), Theory and Practice of Machine Vision (95/100), Introduction to Digital Twin Technology (95/100).

Hebei University of Technology | Micro-major, AI + IoT (Bilingual) 2024.11 – 2025.07

- GPA 4.00/4.00; coursework in embedded system development, intelligent system development, intelligent sensing, edge intelligence, and sensor systems.

University of Arizona, College of Science | Applied physics-related joint training / non-degree coursework 2023.08 – 2025.10

- GPA 3.40/4.00; completed ENGL 106/107/108, PHYS 204, and PHYS 240; A grades in the academic writing sequence.

RESEARCH AND ENGINEERING PROJECTS

Raspberry Pi 5-Based Intelligent Rescue Vehicle System | Main framework, control chain & system integration 2024.07 – 2024.12

- Owned the main control framework and hardware-software integration for a colored-target rescue vehicle while consuming vision outputs from the recognition module; integrated IMU readings, motor/servo/fan/gate control, joystick testing, and task-state transitions.
- Built an end-to-end embedded robotics workflow in Python with Picamera2, RPi.GPIO, gpiozero, and smbus2, connecting perception outputs to actuator control and task-state transitions.
- Awarded Second Prize, Intelligent Rescue Event, 10th Hebei Provincial Engineering Practice and Innovation Competition.

Improved Young's Modulus Measurement System with Remote Preview Module | Co-lead; digital experimental system & host-side software Mar – Sep 2025; awarded Dec 2025

- Co-led a 5-member team improving a traditional physics experiment with manual-reading errors, cumbersome preparation, and limited instructional interactivity; participated in physical-structure principle design and workflow decisions.
- Designed the digital workflow and prototyped the host-side software in C# / WinUI 3, supporting serial data acquisition, dynamic waveform visualization, automatic report generation, and remote preview.
- Converted the experiment into a data-acquisition / processing / visualization / reporting loop; awarded Third Prize, 11th National Undergraduate Physics Experiment Competition (Dec 2025).

Vision Group Hardware Platform, National Undergraduate Smart Car Competition | PCB design & soldering 2023.12 – 2024.07

- Designed and soldered PCB boards for the vision-group smart-car platform; supported hardware debugging and system integration for the vehicle control platform.
- Awarded Second Prize, Regional Contest (Vision Group A), 19th National Undergraduate Smart Car Competition.

Postcard Digital Arrival Platform | Project Lead & Product Architect (BoHack 2026, World Intelligence Expo)

May 2026 (42h)

- Originated the concept from a personal pain point; led product vision and designed a multi-client prototype spanning a Cloudflare Workers backend, React PWA, Android terminal, and Python OCR/AI components during a 42-hour hackathon.
- Prototyped the OCR/AI recognition pipeline in Python (OpenCV image preprocessing + PaddleOCR + local LLM structured extraction); conducted security review of backend TypeScript codebase, identifying critical issues.
- Conducted cultural-tourism market research using national statistics; designed the three-minute pitch, live-demo sequence, and Q&A preparation.

RESEARCH THEMES AND TRANSFERABLE FIT

- Robotics system software: interested in ROS 2/DDS communication, task scheduling, real-time behavior, control-chain latency, and system observability, building on embedded robotics project experience.
- Embedded Linux and device communication: interested in sensor/actuator software stacks, serial/I2C/GPIO communication, and edge-device integration, with hands-on experience in hardware debugging and host-side tooling.
- Adjacent-fit areas: intelligent systems, IoT, digital twins, automated experimental systems, and open-source systems education.

OPEN SOURCE AND TECHNICAL COMMUNITY

AI Innovation Club, Hebei University of Technology | Founder & President

2025.05 – Present

- Founded and operated a campus technical community focused on open source, embedded systems, hardware, and intelligent systems; WeLink community size: 200+.
- Organized OpenCamp, rCore/ArceOS, Linux, and open-source systems learning activities; built knowledge-base and progress-tracking workflows.

OpenCamp & Open-Source Community Outreach

2025.01 – Present

- Outstanding University Volunteer, OpenCamp Open Source Training Program, Tsinghua University OS Laboratory; public award page: opencamp.cn/awards/2025
- Selected for the first cohort of the Chance Foundation Open Source Evangelist Program and received the Rising Star recognition; participated in open-source events, hackathons, and campus technical outreach.

HONORS AND AWARDS

- 2025 | Second Prize, Intelligent Rescue Event, 10th Hebei Provincial Engineering Practice and Innovation Competition
- 2024 | Second Prize, Regional Contest (Vision Group A), 19th National Undergraduate Smart Car Competition
- 2025 | Third Prize, 11th National Undergraduate Physics Experiment Competition
- 2025 | Outstanding University Volunteer, OpenCamp Open Source Training Program, Tsinghua University OS Laboratory
- 2025 | Rising Star, Open Source Evangelist Program, Chance Foundation

TECHNICAL SKILLS

- Programming used in projects: Python, C/C++, C#/ .NET. Coursework / active learning: Rust, R. Systems & Embedded: Linux, Git, Raspberry Pi, GPIO, I2C, serial communication, IMU, actuator control. Robotics & Vision: OpenCV, Picamera2, embedded vision interfaces, hardware-software integration. Data & Tools: NumPy, experimental data processing, visualization, documentation, and open-source collaboration.